Missing Clinical Covariates in a Small Regression Model

Complete Case Analysis versus Multiple Imputation

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Abstract

Clinical regression models are often fit with partially missing baseline covariates such as laboratory values, body mass index, and smoking status. A common response is complete-case analysis, which discards patients with any missing covariate information and proceeds as though the retained subset still represents the target analysis population. We study a clinical logistic regression with five baseline predictors and three partially observed covariates under increasing missingness severity. Complete-case analysis is compared with multiple imputation by chained equations under MCAR, MAR, and an MNAR-style stress scenario. Performance is evaluated using coefficient bias, root mean squared error, nominal 95% confidence interval coverage, and retained sample size. Under MCAR, complete-case analysis primarily reduces precision through sample loss. Once missingness depends on observed patient characteristics or routine workflow patterns, however, complete-case analysis can also distort inference by changing who remains in the data. The main practical lesson is that dropping incomplete records is not merely a data-cleaning step; it can silently redefine the study population and alter the coefficient being estimated.

1. Applied Problem

Suppose an analyst wants to estimate the association between a baseline inflammation-related laboratory value and a 30-day adverse event outcome using a multivariable clinical regression model. The available baseline predictors are age, comorbidity burden, the laboratory value of interest, body mass index, and current smoking status.

In practice, some of these baseline covariates are often missing. Laboratory measurements may be absent because they were ordered selectively, body mass index may be unrecorded because intake documentation was incomplete, and smoking status may be missing because history-taking was inconsistent. A common response is to restrict the analysis to patients with complete information on all covariates and fit the regression only on that subset.

At first glance, this approach seems appealing because it is simple to implement and requires no explicit model for the missing-data process. The difficulty is that once covariate missingness is related to patient characteristics or routine workflow patterns, the patients retained for analysis may no longer represent the full target population. In that setting, deleting incomplete records is not merely a reduction in sample size; it can also change the population to which the fitted regression effectively applies.

2. Failure Mode

The central failure mode is structured covariate missingness.

In clinical data, missingness often reflects routine care and documentation processes rather than harmless random loss. A laboratory value may be missing because testing was ordered selectively, body mass index may be absent because intake was incomplete, and smoking status may be unrecorded because history-taking varied across patients. When missingness follows these kinds of patterns, the patients with fully observed covariates are no longer a random subset of the original cohort.

This is where complete-case analysis begins to fail. By discarding all observations with any missing covariate, it can replace the intended analysis population with a filtered subset defined partly by care pathways and documentation patterns. The fitted regression may still converge and produce standard output, but the resulting estimates may describe associations in the wrong population.

The practical claim is simple: once covariate missingness tracks patient characteristics or routine workflow patterns, complete-case analysis is no longer just a loss of sample size.

3. Comparison Setup

3.1 Outcome and predictors

The outcome is a binary indicator of a 30-day adverse event. The analysis model is a logistic regression with five baseline covariates: age, comorbidity burden, an inflammation-related laboratory value, body mass index, and current smoking status.

To create an application-like setting, the continuous predictors are first generated on raw scales and then standardized for modeling. Age, the laboratory value, and body mass index enter the fitted regression as standardized continuous predictors. Comorbidity burden is coded as an ordinal variable with levels 0, 1, and 2, and smoking is a binary current-smoking indicator.

Standardizing the continuous predictors places their coefficients on a common scale, so each coefficient represents the change in log-odds associated with a one-standard-deviation increase in that predictor, holding the others fixed.

The parameter of primary interest is the regression coefficient for the laboratory value.

3.2 Data-generating model

For subject i, the outcome is generated from the logistic model

$$\text{logit}\mathrm{P}(Y_i = 1 \mid X_i) = \beta_0 + \beta_{\text{age}}\,\text{age}_i + \beta_{\text{comorb}}\text{comorbidity}_i + \beta_{\text{lab}}\,\text{lab}_i + \beta_{\text{bmi}}\,\text{bmi}_i + \beta_{\text{smoke}}\text{smoking}_i.$$

The coefficients are chosen to produce moderate signal strength and clinically plausible directions of association:

Table 1: Coefficient values used in the data-generating logistic model.

Term	Symbol	Coefficient
Intercept	β_0	-1.20
Age	β_{age}	0.60
Comorbidity	β_{comorb}	0.50
Laboratory value	β_{lab}	0.80
Body mass index	β_{bmi}	0.35
Current smoking	β_{smoke}	0.55

The simulation first generates correlated latent covariates using a general-purpose data-generating function, then rescales them to application-like raw values and standardizes the continuous predictors for outcome generation and estimation. This produces a controlled setting in which age, laboratory value, and body mass index are correlated rather than independent.

3.3 Missingness structure

Three covariates are allowed to be partially missing: the laboratory value, body mass index, and smoking status. Missingness severity is varied across mild, moderate, and strong settings.

Three missingness regimes are considered.

MCAR. Missingness is generated independently of the covariates and outcome, using fixed variable-specific probabilities. This serves as a benign benchmark in which complete-case analysis is expected to lose information primarily through reduced sample size rather than through systematic distortion of the retained cohort.

MAR. Missingness depends only on predictors that remain observed in the analysis. In the simulation, missingness probabilities are functions of age and comorbidity. The intended clinical story is routine workflow missingness: the laboratory value is more likely to be missing in lower-risk patients because testing is ordered selectively; body mass index is more likely to be missing in older or more comorbid patients because intake documentation is less complete; and smoking status is also more likely to be missing in older or more comorbid patients, though at a lower baseline rate. Increasing severity strengthens the dependence of missingness on these observed patient characteristics.

MNAR-style stress case. Missingness depends on observed age and comorbidity, as in the MAR setting, but also on the covariate's own full-data value before missingness is imposed. In particular, laboratory missingness is higher for lower underlying laboratory values, body mass index missingness is higher for lower underlying BMI values, and smoking-status missingness is higher among smokers. This produces a setting in which the missingness process is not fully explained by variables that remain observed in the final dataset. The purpose of this design is not to give a full treatment of MNAR identification, but to stress both methods by introducing dependence on information unavailable to the observed-data model.

3.4 Methods compared

Two methods are compared.

Complete-case analysis: fit the logistic regression only on observations with no missing covariates.

Multiple imputation: use chained equations to impute missing covariates, then fit and pool the logistic regression estimates across completed datasets.

Multiple imputation is implemented using predictive mean matching for the continuous covariates and logistic regression for smoking status. The outcome was included as a predictor in the imputation model but was not itself imputed. After imputation, the logistic regression is fit within each completed dataset and the resulting estimates are combined using Rubin's rules.

3.5 Performance metrics

The primary performance measures are bias, root mean squared error, nominal 95% confidence interval coverage, and average retained sample size under complete-case analysis, all evaluated for the regression coefficient of the laboratory value.

A descriptive compositional diagnostic is also reported: the shift in mean true risk score between the full cohort and the retained complete-case subset, where the risk score is defined by the linear predictor used in the data-generating model.

4. Results

The simulation results support a simple practical pattern. Under MCAR, complete-case analysis mainly loses efficiency as missingness increases. Under MAR, complete-case analysis increasingly alters the composition of the retained cohort, with growing risk-score distortion and modest degradation in coefficient performance. Under the MNAR-style stress case, both methods operate under misspecification risk, but complete-case analysis still discards a large fraction of the sample and remains vulnerable to substantial compositional shift.

4.1 MCAR

Under MCAR, complete-case analysis behaves largely as a deletion-based efficiency loss. The mean complete-case estimate for the laboratory coefficient remains close to the true value of 0.8 across severity levels, with bias ranging from about 0.009 to 0.029. The main deterioration is in precision: RMSE rises from 0.109 under mild MCAR to 0.156 under strong MCAR as the average retained sample size falls from about 787 to 399 observations.

The compositional diagnostic confirms that the retained complete-case subset remains very similar to the full cohort under MCAR. The average risk shift is essentially zero at all severity levels, indicating that deletion is not materially changing who remains in the analysis.

Multiple imputation is more efficient in this setting and also shows slightly smaller bias than complete-case analysis across the MCAR scenarios. Its bias ranges from about 0.007 to 0.015, compared with about 0.009 to 0.029 for complete-case analysis, and its RMSE ranges from 0.101 to 0.120, consistently below the corresponding complete-case values. Nominal 95% interval coverage remains similar for the two methods. In this benchmark setting, the main penalty of complete-case analysis is therefore reduced information rather than strong target-population distortion.

Table 2: MCAR results across missingness severity.

Severity	CC Bias	CC RMSE	CC Coverage	CC Retained	MI Bias	MI RMSE	MI Coverage	CC Risk Shift
mild	0.011	0.109	0.944	0.787	0.008	0.101	0.948	0.000
moderate	0.009	0.118	0.948	0.612	0.007	0.105	0.948	0.000
strong	0.029	0.156	0.950	0.399	0.015	0.120	0.950	-0.003

4.2 MAR

Under MAR, complete-case analysis no longer acts as a harmless sample-size reduction. As missingness severity increases, the retained complete-case sample becomes increasingly different from the full cohort. The mean risk shift becomes progressively more negative, moving from about -0.035 under mild MAR to -0.217 under strong MAR. This indicates that deletion is selectively removing higher-risk or differently composed patients rather than preserving a representative subset of the original cohort.

This compositional distortion is accompanied by weaker complete-case performance. The complete-case laboratory coefficient remains only modestly biased in absolute terms, but bias and RMSE both increase with severity: bias rises from about 0.011 to 0.021, and RMSE rises from 0.117 to 0.130. The average retained sample size also declines from about 648 to 533 observations.

Multiple imputation performs more favorably across the MAR settings. Its bias remains small and is modestly improved relative to complete-case analysis, and its RMSE is consistently lower, ranging from about 0.097 to 0.102. Multiple imputation also avoids the same degree of effective sample loss because all 1000 observations remain in the analysis after imputation. In this setting, the main practical problem is not dramatic coefficient bias alone, but the combination of sample loss, reduced efficiency, and increasing divergence between the retained complete-case subset and the intended target cohort.

Table 3: MAR results across missingness severity.

Severity	CC Bias	CC RMSE	CC Coverage	CC Retained	MI Bias	MI RMSE	MI Coverage	CC Risk Shift
mild	0.011	0.117	0.952	0.648	0.010	0.102	0.942	-0.035
moderate	0.015	0.124	0.938	0.596	0.006	0.100	0.936	-0.118
strong	0.020	0.130	0.958	0.533	0.011	0.097	0.950	-0.217

4.3 MNAR-style stress case

Under the MNAR-style stress case, neither method should be treated as uniformly correct, because the missingness mechanism depends partly on information unavailable in the observed-data model. This is exactly the setting in which multiple imputation loses the protection it has under a correctly specified MAR mechanism.

Even so, complete-case analysis is not rescued by the fact that imputation is also under strain. Across the MNAR-style scenarios, complete-case analysis retains only about 475 to 549 observations on average, implying that roughly half the sample is discarded. The compositional distortion is also substantial, with positive mean risk shifts of about 0.260, 0.198, and 0.126 across the mild, moderate, and strong settings. The retained complete-case subset is therefore again meaningfully different from the full cohort, though here the shift is in the opposite direction from the MAR setting. The complete-case risk shift is not monotone in severity, which reflects the fact that stronger missingness does not necessarily translate into a larger mean-risk difference in the retained subset when deletion depends on several covariates simultaneously.

In this stress case, multiple imputation remains competitive and is usually more efficient. Its RMSE ranges from about 0.096 to 0.112, compared with 0.113 to 0.137 for complete-case analysis. Bias is small for both methods, and interval coverage is generally acceptable for both, with some scenarios favoring one method slightly over the other. The main interpretive lesson is therefore not that multiple imputation solves MNAR, but that complete-case deletion remains a fragile strategy even when imputation is also operating outside its ideal assumptions.

Table 4: MNAR-style stress-case results across missingness severity.

Severity	CC Bias	CC RMSE	CC Coverage	CC Retained	MI Bias	MI RMSE	MI Coverage	CC Risk Shift
mild	0.012	0.133	0.932	0.549	0.014	0.112	0.940	0.260
moderate	0.007	0.113	0.982	0.518	0.006	0.096	0.972	0.198
strong	0.011	0.137	0.942	0.475	0.012	0.104	0.946	0.126

5. Practical Consequences

If this issue is ignored, the fitted model may end up describing associations in a filtered subset of patients rather than in the intended clinical population.

The practical failure is not limited to loss of sample size. Under structured missingness, complete-case analysis can change who remains in the analysis, reduce precision, and shift the effective target population even when the regression converges without warning. In the MAR settings examined here, the complete-case subset became increasingly different from the full cohort as missingness severity increased. In the MNAR-style stress setting, substantial compositional distortion remained even though multiple imputation was also operating under strain.

This matters because analysts often judge missing-data damage only by how many records were deleted. That is incomplete. The deeper question is whether deletion has altered the composition of the analytic sample enough to change the meaning of the fitted coefficient.

6. What Practitioners Should Notice

Before treating complete-case analysis as harmless, practitioners should look for practical warning signs.

- Missingness rates differ across age, comorbidity, or other observed risk strata.
- The complete-case subset looks materially different from the full analytic cohort.
- Coefficient estimates or standard errors change meaningfully after multiple imputation.
- Missingness indicators are predictable from observed covariates.
- Deletion removes substantially more information than the marginal missingness percentages suggest when several covariates are partially observed.

These signs do not by themselves prove that complete-case analysis is invalid, but they do indicate that it is unlikely to be innocuous.

7. Discussion

This analysis is intentionally narrow. It is not a full treatment of missing-data theory, sensitivity analysis, inverse probability weighting, or MNAR identification. Its purpose is to demonstrate one practical point in a clinical regression setting with several partially observed baseline covariates.

The results support a simple conclusion. Under MCAR, complete-case analysis behaves mostly as a precision loss: the retained sample remains compositionally similar to the full cohort, bias stays small, and the main penalty is reduced efficiency as missingness increases. Under MAR, complete-case analysis becomes more problematic because deletion increasingly filters the cohort, producing both efficiency loss and growing divergence between the retained subset and the intended target population. Under the MNAR-style stress case, neither method should be treated as universally correct, but complete-case analysis is not rescued by that fact: it still discards a large fraction of the sample and can still induce substantial compositional distortion.

Multiple imputation is therefore not presented here as a cure-all. Its role is more modest and more practical. In these simulations, it generally preserved estimation more effectively than complete-case analysis under MAR, remained competitive under the MNAR-style stress case, and avoided the same degree of effective sample loss throughout. In this setting, it is the stronger default.

Code used to generate the simulations, impose missingness, fit both analysis strategies, and summarize performance is provided in the accompanying repository.

References

Rubin, D. B. (1976). Inference and missing data. Biometrika, 63(3), 581–592. https://doi.org/10.1093/biomet/63.3.581

van Buuren, S. (2018). Flexible imputation of missing data (2nd ed.). Chapman and Hall/CRC. https://doi.org/10.1201/9780429492259

van Buuren, S., & Groothuis-Oudshoorn, K. (2011). mice: Multivariate imputation by chained equations in R. Journal of Statistical Software, 45(3), 1–67. https://doi.org/10.18637/jss.v045.i03

Appendix

R Code

```r
library(tidyverse)
library(parallel)
library(mice)
library(knitr)

setwd("/Users/subspace/Desktop/Short Projects/Data Generation/")

source("generate_sim_data.R")
source("missing_covariate_note_helpers.R")

# Number of worker processes to use.
# Use a conservative number to avoid memory pressure from MICE.
n_cores <- min(4, max(1, parallel::detectCores() - 1))
```

```r
# Scenario grid
settings <- expand.grid(
  mechanism = c("MCAR", "MAR", "MNAR"),
  severity = c("mild", "moderate", "strong"),
  stringsAsFactors = FALSE
)

cl <- parallel::makeCluster(n_cores)
on.exit(parallel::stopCluster(cl), add = TRUE)

# Load needed packages and source files on each worker
parallel::clusterEvalQ(cl, {
  library(mice)
  library(dplyr)
  library(purrr)
  library(tibble)
  NULL
})

parallel::clusterExport(
  cl,
  varlist = c("settings"),
  envir = environment()
)

parallel::clusterEvalQ(cl, {
  source("generate_sim_data.R")
  source("missing_covariate_note_helpers.R")
  NULL
})

# Run one full scenario per worker task

(start <- Sys.time())

scenario_results <- parallel::parLapply(cl, seq_len(nrow(settings)), function(s) {
  mech <- settings$mechanism[s]
  sev <- settings$severity[s]

  run_simulation(
    n_sims = 500,
    n = 1000,
    mechanisms = mech,
    severities = sev,
    mi_m = 20,
    base_seed = 20260327 + 10000 * s
  )
})

sim_res <- dplyr::bind_rows(scenario_results)

# setwd("/Users/subspace/Desktop/Short Projects/Healthcare/")
# save(sim_res, file = "sim_res.RData")
```

```r
end <- Sys.time()
print(end - start)
```

Code Validation

The following code tests `missing_covariate_note_helpers.R` to verify that the simulation, missingness mechanisms, estimation routines, and summary outputs behave as intended.

Code Validation Summary

These validation checks indicate that `missing_covariate_note_helpers.R` is behaving consistently with the design of the note.

The baseline data-generation check shows that the simulated cohort has plausible raw predictor scales, centered standardized covariates, and a nondegenerate binary outcome. In the representative run shown here, the event prevalence is approximately 0.40, which is appropriate for the intended small clinical logistic-regression setting.

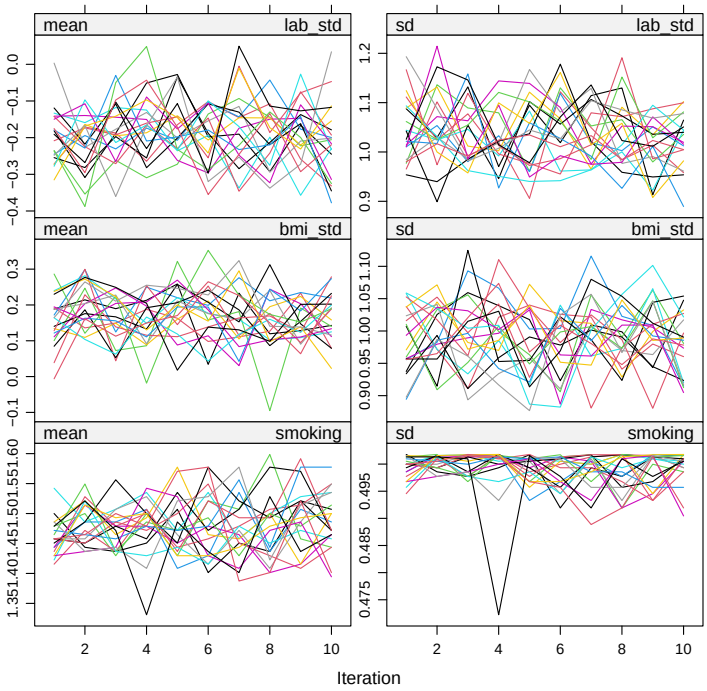
The missingness checks confirm that the three mechanisms are being imposed distinctly. Under MCAR, missingness rates are set independently of the covariates and primarily reduce the available sample size. Under MAR, missingness varies with observed patient features such as age and comorbidity, producing differential retention across risk strata. Under the MNAR-style stress case, missingness depends partly on the latent covariate values themselves, leading to lower complete-case retention and stronger compositional distortion than under MAR. The synchronization checks also confirm that when lab or BMI is set to missing on the raw scale, the corresponding standardized version is set to missing in the same rows.

The single-dataset estimation checks show that both analysis routines run successfully. Complete-case analysis fits on the retained subset only, while multiple imputation uses all rows and returns pooled coefficient estimates and interval summaries without error. The one-replicate pipeline check further confirms that a full scenario run returns the expected output structure, including coefficient estimates, interval limits, missingness rates, retained sample proportion, and the complete-case risk-shift diagnostic.

The repeated-simulation check shows that the helper file produces the intended qualitative behavior across scenarios. Under MCAR, complete-case analysis remains approximately unbiased and mainly loses precision through sample-size reduction. Under MAR, complete-case retention falls and the retained subset differs more noticeably from the full cohort, with corresponding degradation in complete-case performance. Under the MNAR-style stress case, complete-case retention falls further and the retained subset shows more pronounced risk-score distortion. Multiple imputation still uses the full sample and often remains competitive, but it is no longer protected by a correctly specified observed-data missingness mechanism, so its performance should not be interpreted as uniformly superior.

Taken together, these checks support using `missing_covariate_note_helpers.R` for the full simulation study in the note. The data-generation, missingness, estimation, and summary components appear to be functioning as intended.

Multiple-imputation diagnostic check



To assess whether the chained-equations algorithm appeared to mix adequately, we examined trace plots of the mean and standard deviation of the imputed values across iterations. We looked for persistent drift, separation among chains, or other signs that between-chain variability remained systematically larger than within-chain variability. The MICE trace plots were consistent with adequate mixing: the imputation streams fluctuated around stable levels without obvious drift or persistent chain separation.